Finding min and max concurrently

Suppose you have an array of n **distinct** numbers (or other comparable element). In the ICS 30-series, you learned how to find the min or max of such an array. Suppose you wanted to find *both* – the min and max.

One way to do this would be to find the min; this takes $n-1$ comparisons. You could then output and delete the min element and find the max of what remains, taking $n-2$ comparisons, for a total of $2n-3$ comparisons.

We are going to try to find a way to find both the min and max using *strictly fewer than* $2n-3$ *comparisons*, Note that we are measuring the actual number of comparisons, not the growth rate of your function.

If you are having trouble starting, you may assume n to be odd or even (your choice).

To do this, let's consider what the worst-case will look like. We are going to examine an **adversary argument**: that is, we have an opponent who is going to examine our code and set up the input to cause us to use as many comparisons as they are able to force.

As shorthand, when we compare a and b, if $a < b$, we say that a "lost" the comparison/competition and b "won."

Every time we find out that a given element x is not the maximum, that is one unit of information we learned about the correct output. We similarly gain one unit of information every time one element is discovered to not be the minimum. In total, we must learn $2n-2$ units of information: each element other than the maximum must be discovered to not be the maximum, and similar for the minimum.

Question 1. Suppose I am comparing two array elements, a and b. What should the adversary do if neither a nor b has been compared before?

Question 2. Suppose I am comparing two array elements, a and b. What should the adversary do if exactly one of a and b has participated in a comparison before?

Question 3. How many times can we gain two units of information with one comparison?

Question 4. What is the fewest comparisons any solution can guarantee they use? Give an algorithm that uses that many comparisons.

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